

SLS Capstone Report

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Executive Summary:

Sim's Lifecycle Services (SLS) previously allocated approximately 80 cumulative labor hours per reporting cycle to manually extract Product Carbon Footprint (PCF) data from manufacturer PDFs. To address this operational bottleneck, a fully automated, multimodal vision pipeline was developed within the OpenAI ecosystem. This solution bypasses traditional optical character recognition to preserve the semantic mapping of complex graphical structures, including split-legend pie charts. The architecture employs a token-optimized, three-tier escalation protocol, defaulting to high-efficiency models such as GPT-4o-mini and escalating to advanced models only for visual edge cases. Automated validation layers support anomaly detection and algorithmic normalization. This deployment increased extraction velocity by over 98%, reducing the 80-hour manual workflow to less than one hour of unattended compute time. By processing over 1,300 products at a total API cost of \$31.85, the system achieves estimated quarterly savings of \$2,350 and 75 labor hours, enabling SLS to transition from manual data transcription to a scalable, high-margin data strategy.

Introduction:

Sim's Lifecycle Services (SLS) has been manually updating its SLS Calculator V8_Carbon Avoidance Report. This Excel workbook contains all the information needed to compute carbon footprint emissions for client communication every six months. Their current system involves searching each company's website for Product Carbon Footprint (PCF) files and manually copying relevant information into the workbook. This system is temporally costly, which is why we developed a pipeline to streamline the PCF data collection process. This pipeline aimed to improve data collection efficiency while maintaining data integrity.

Text Extraction:

Modern file data harvesting techniques increasingly leverage automated text extraction tools to parse and convert unstructured documents into machine-readable text (Zhang et al., 2024). The need for such tools arises from the rapid growth of unstructured digital data, including documents, images, and reports, which cannot be effectively processed with traditional methods for structured data (Philippovich & Philippovich, 2025). Automated text extraction typically follows a multistep process involving localization, segmentation, and binarization. Localization identifies the precise position of textual elements within a document, while segmentation separates the text from surrounding background or non-text elements. Binarization then converts the image into a simplified binary format to facilitate further processing, after which optical character recognition (OCR) techniques convert detected characters into machine-readable ASCII text (Kaundilya et al., 2019; Mittal & Garg, 2020). In addition to

image-based processing, modern extraction systems must address the structural complexity of digital document formats such as PDF, which often contain mixed narrative text, metadata, and semi-structured elements such as tables and figures. Because these documents frequently lack explicit structural markup, automated systems must infer logical document structure from layout, typographical cues, and spatial relationships between text elements (Hassan & Baumgartner, 2005).

Recent research emphasizes hybrid extraction approaches that combine document parsing libraries, optical character recognition, and natural language processing techniques to preserve document structure while extracting textual content. These approaches improve the identification of document components such as headings, tables, and paragraphs and enable the extraction of information into structured datasets suitable for downstream analysis (Philippovich & Philippovich, 2025). Empirical evidence further demonstrates the practical value of these systems. For example, Vist et al. (2025) show that automated extraction tools can efficiently convert information from unstructured PDF documents into structured datasets, significantly enhancing systematic literature review workflows. Advances in machine learning have further strengthened these capabilities. Deep learning models, particularly transformer-based architectures, enable context-aware parsing of complex textual data and support scalable, accurate extraction from large document collections (Silva & Silva, 2021).

Beyond improvements in extraction algorithms, operationalizing these techniques at scale relies heavily on application programming interfaces (APIs), which enable programmatic access to document repositories and facilitate integrating extraction capabilities into broader automated data-harvesting systems. APIs act as intermediaries, allowing systems to communicate by sending documents or extraction requests to processing services and returning structured, machine-readable outputs that can be seamlessly incorporated into automated data pipelines.

Application Programming Interfaces:

Application programming interfaces (APIs) are widely recognized as a key mechanism for enabling communication between software systems, allowing applications to exchange data and access external services through standardized requests and responses (Wu et al., 2020). Prior research demonstrates that APIs enable the integration of external services, including artificial intelligence models and data platforms, into automated computational workflows (Perron et al., 2025; Ye et al., 2024; Galli et al., 2025). In document analysis systems, APIs enable documents or text inputs to be transmitted to externally hosted models that perform tasks such as information extraction, summarization, and classification, returning structured outputs suitable for downstream analysis (Moreira-Filho et al., 2025). Recent studies further demonstrate that automated data extraction pipelines can leverage large language models accessed through APIs to process full-text research articles and extract structured variables for systematic reviews. For example, Khan and colleagues (2025) developed a workflow that transmits text and images from clinical trial publications to GPT-4 and Claude models via APIs, enabling the automated extraction of structured data from PDF documents with high accuracy, directly supporting the

potential of API-based language model involvement in expedited, scalable systematic review workflows.

Currently Available Tools on the Market:

Commercial text extraction systems are increasingly delivered as cloud-based services that expose document analysis capabilities through application programming interfaces (APIs). These platforms are commonly described as document intelligence systems that combine optical character recognition (OCR), layout analysis, and structured data extraction within scalable cloud infrastructures. Major providers, including Amazon Web Services, Google Cloud, and Microsoft Azure, offer APIs capable of extracting text, detecting document structures such as tables and form fields, and returning machine-readable outputs designed for integration into automated data processing pipelines (Amazon Web Services, 2024; Google Cloud, 2024; Microsoft, 2024). For example, Amazon Textract applies machine learning to identify text, tables, and key-value pairs in scanned documents and provides structured outputs that can be incorporated into downstream analytical workflows (Shree et al., 2025). These services are typically accessed programmatically via REST APIs, enabling developers to integrate document-processing functionality directly into software systems and large-scale data pipelines.

Empirical research supports the growing adoption of these cloud-based document intelligence platforms. Benchmarking experiments comparing widely used OCR tools such as Tesseract with server-based systems, including Amazon Textract and Google Document AI, show that cloud-hosted processors generally achieve higher accuracy and greater robustness when processing noisy or degraded documents (Hegghammer, 2021). These improvements reflect broader advances in machine-learning-driven document analysis and in the scalability of cloud infrastructure. At the same time, the capabilities of text extraction systems are expanding beyond traditional OCR. Recent work demonstrates that AI-driven pipelines combining multimodal language models and OCR services can automate large portions of document transcription workflows. For example, experiments integrating GPT-4 vision models with Google Cloud Vision show that hybrid AI extraction pipelines can automatically process large image collections while significantly reducing both costs and manual labor requirements (Robotko, 2025).

More broadly, the rapid expansion of machine learning APIs has enabled developers to incorporate advanced extraction and analysis capabilities without training custom models. ML APIs expose pre-trained models through standardized interfaces, enabling applications to perform tasks such as image recognition, text extraction, and language processing via simple service calls (Jiang et al., 2025). Because these APIs abstract the complexity of model training and infrastructure management, they have become foundational components in modern document processing architectures. As a result, contemporary text extraction systems are increasingly implemented as integrated pipelines that combine OCR services, document intelligence APIs, and large language models to convert unstructured documents into structured datasets suitable for large-scale computational analysis.

Collectively, the literature demonstrates that automated text extraction technologies, API-based processing, and cloud-based document intelligence platforms provide the necessary infrastructure for building automated document data collection systems. Therefore, we propose developing a text-extraction pipeline that uses an API to automate the collection and parsing of Product Carbon Footprint (PCF) data from manufacturer reports, combining vision-enabled large language models with tiered extraction and validation mechanisms to populate the SLS Calculator.

Methods:

Experimental Setup and Enterprise Alignment:

To establish a performance baseline, the initial pipeline was tested using a standardized dataset of Dell PCF reports. Preliminary evaluations compared commercial large language models (LLMs) across the OpenAI, Google, and Anthropic ecosystems, assessing each for execution latency, token efficiency, and extraction precision. While early prototyping showed high parsing accuracy with alternative models, the final architecture was optimized exclusively for the OpenAI ecosystem to ensure strict alignment with enterprise API provisioning and client infrastructure requirements.

Phase I: The Hybrid OCR Hypothesis:

The first architectural iteration employed a hybrid text-extraction system, pairing Azure Optical Character Recognition (OCR) with a text-only LLM. While this approach successfully parsed standard tabular data, it systematically failed on complex, semi-structured graphical elements. Traditional OCR protocols decoupled the semantic relationships within complex visual formats, such as split-legend pie charts. Consequently, the downstream text-only models were unable to accurately map key lifecycle values for the Manufacturing, Transportation, Use, and End-of-Life stages. To compensate, an intermediary Python script was deployed to prompt the LLM to identify the missing variables in the raw text, but extraction reliability remained suboptimal.

Phase II: The Multimodal Vision Pivot:

Late-stage validation revealed that relying on OCR for visually dominant PCF reports introduced unacceptable error rates. To resolve the semantic decoupling inherent in the OCR pipeline, the architecture was pivoted to a strictly multimodal, vision-based approach. By discarding Azure OCR entirely, the pipeline was re-engineered to encode target PDF pages as high-resolution base64 images—a text representation of binary data. Feeding raw visual data directly into OpenAI's vision models preserved all spatial relationships and graphical context, thereby drastically improving the precision of extracting complex charts and graphs.

Phase III: Pre-Processing and Pipeline Efficiency:

To operationalize this vision architecture at scale, the production pipeline uses a multi-tiered, resource-optimized approach to minimize API token usage. The backend orchestration begins by deduplicating product SKUs mapped to the same PDF URL, thereby preventing redundant API calls. Following deduplication, a deterministic algorithm isolates target data pages via keyword heuristics, identifying key Life Cycle Assessment terms such as "kg CO2e" or "%". This filtering mechanism automatically skips verbose, non-technical pages to avoid model hallucination and conserve computational resources.

Phase III: The Three-Tier Escalation Protocol:

A three-tier escalation protocol governs the automated extraction process:

- **Tier 1 (Standard Baseline):** The isolated target page is converted to a base64-encoded image and processed via a high-efficiency OpenAI model (e.g., GPT-4o-mini) at a baseline resolution of 200 DPI.
- **Tier 2 (Context Expansion):** If critical carbon fields return null values during the Tier 1 pass, the system automatically expands its search window to three adjacent pages and increases the image input resolution to 320 DPI.
- **Tier 3 (Frontier Escalation):** Persistent null values trigger an immediate escalation to OpenAI's premium frontier model (e.g., GPT-4o). This tier is reserved solely for handling edge-case visual complexities, ensuring that high-cost tokens are reserved exclusively for structurally anomalous documents.

Quality Assurance and Statistical Anomaly Detection:

To guarantee dataset integrity and mitigate LLM hallucination, the pipeline integrates three automated validation layers:

- **Algorithmic Normalization:** The system applies deterministic mathematical formulas to impute missing values. If the model successfully extracts total emissions and lifecycle percentages but omits the raw mass figures, the backend autonomously computes and populates the missing variables.
- **Missingness Tracking:** Null values that bypass the normalization layer are programmatically flagged, generating a clear audit trail for rapid user review and manual gap remediation.
- **Percentile Outlier Detection:** Extracted values are dynamically benchmarked against intra-category distributions. Metrics falling within the top or bottom 5% percentiles of their respective product category are automatically isolated for human validation, significantly reducing the propagation of extraction errors into the final database.

Results: Operational Impact and Efficiency Gains

Operational Velocity and Labor Reduction:

Deploying the automated OpenAI data pipeline fundamentally transformed the PCF collection workflow. Previously, manual data harvesting required about 80 cumulative labor hours per reporting cycle. Staff had to locate, read, and transcribe lifecycle metrics. The new multimodal architecture compresses this entire process into less than one hour of unattended compute time. By accelerating extraction by over 98%, the pipeline removes a key bottleneck. SLS can now redirect valuable staff hours to higher-level data analysis and client strategy.

Financial Scalability and Token Efficiency:

The multi-tiered escalation protocol minimized computational overhead while maintaining data integrity. High-efficiency baseline models, such as GPT-4o-mini, handled extractions by default. The total cost of running the scraping pipeline through more than 1300 products was \$31.85. It is estimated that this would save SLS approximately \$2,350 per year. It is also estimated that running the pipeline would save our client more than 75 hours of labor, as the client stated it takes approximately 2 full weeks of work per year and that the scraper can run unattended.

Data Standardization and Integrity Assurance:

Shifting from manual data entry to an automated, centralized database eliminated human transcription errors. Error-reduction metrics were not tracked empirically during the transition. However, validation layers in the pipeline consistently improved data integrity. Algorithmic normalization and missingness tracking flagged anomalies and nulls, generating clear audit trails before data ingestion. The pipeline ensures the SLS Calculator has only standardized, high-fidelity metrics. This greatly reduces the need for manual oversight.

Discussion and Next Steps:

Strategic Value and Operational Realignment:

The deployment of this automated pipeline transitions Sim's Lifecycle Services (SLS) from manual data processing to a high-margin data strategy. By fully automating PCF extraction, the architecture eliminates significant operational overhead, reallocating staff hours toward advanced analytics and strategic client advisory. The auto-updating repository ensures the SLS Calculator remains standardized and continuously deployment-ready, providing a scalable foundation for expanding the firm's sustainability offerings.

Architectural Constraints and Edge Cases:

While the multimodal vision architecture effectively parses standard PCF reports, extraction fidelity occasionally degrades when processing highly idiosyncratic visual elements. Specifically, edge-case graphical structures—such as pie charts with spatially decoupled legends—can disrupt the model's semantic mapping. Although the pipeline's automated anomaly detection successfully isolates these instances for manual review, fully eliminating human-in-the-loop dependencies will require continued advancements in multimodal spatial reasoning.

Phase II: Horizontal Scaling and Market Expansion

With the foundational architecture, tiered escalation protocols, and validation heuristics successfully operationalized for the initial cohort of four manufacturers, the primary directive for future development is horizontal scaling. The next phase of deployment must expand the pipeline's extraction capabilities to encompass a broader spectrum of hardware producers. Onboarding new manufacturers will require iterative tuning of the deterministic filtering algorithms to accommodate proprietary PDF layouts, and scaling this infrastructure will significantly increase the repository's data density, ultimately equipping SLS with a comprehensive, industry-wide Carbon Avoidance asset.

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Appendix:

API Pricing Table:

Assumptions: 1,300 products; 3 pages per product; 2,500 input tokens per page; 500 output tokens per page.

Cost Component	Assumption / Formula	Token Volume	Rate	Estimated Cost
Products	Given	1,300 products	—	—
Pages per product	Given	3 pages/product	—	—
Total pages processed	$1,300 \times 3$	3,900 pages	—	—
Input tokens per page	Given	2,500 input tokens	—	—
Total input tokens	$3,900 \times 2,500$	9,750,000	\$1.75 / 1M tokens	\$17.06
Output tokens per page	Given	500 output tokens	—	—
Total output tokens	$3,900 \times 500$	1,950,000	\$14.00 / 1M tokens	\$27.30
Total estimated API cost	Input cost + output cost	11,700,000 total tokens	—	\$44.36

Note. Estimate assumes one API call per page, no cached-input discounts, no Batch API discount, and no additional tool, file, image, or retrieval costs. Pricing is based on the GPT-5.2 text-token rates used in the estimate: \$1.75 per 1M input tokens and \$14.00 per 1M output tokens.

-This table was generated by ChatGPT

Assuming manual PCF data entry requires 80 hours at \$30 (low estimate) per hour, the estimated manual processing cost per reporting cycle is \$2,400. Using the OpenAI API instead would cost approximately \$44.36, resulting in an estimated cost savings of \$2,355.64 per cycle.